

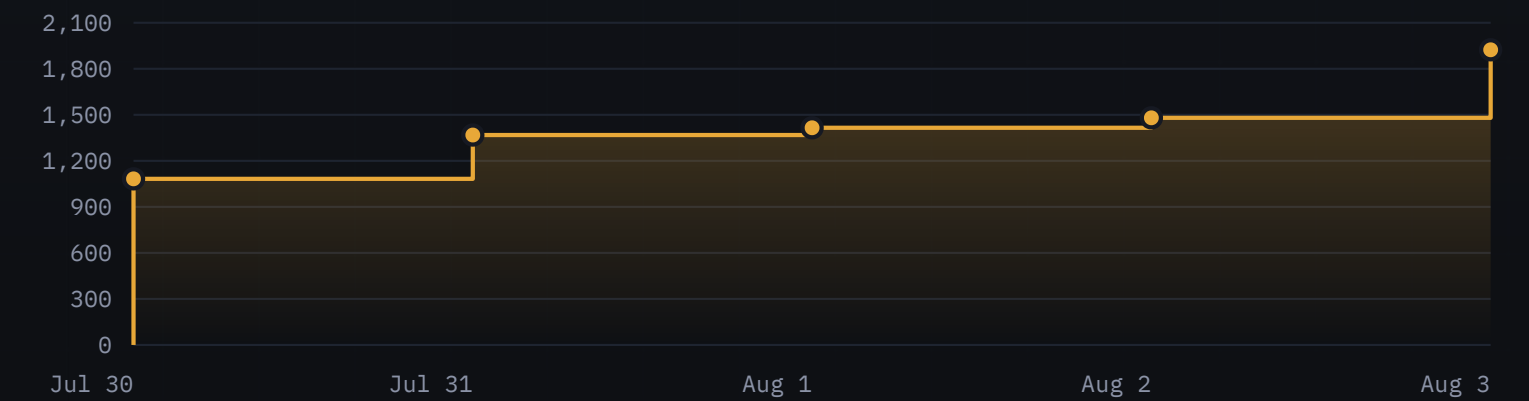
Coldcard Hack Tracker

Seed-entropy sweeps since July 2026

TOTAL STOLEN

1,923.80748125 BTC
\$125,109,048

Known waves only · total most likely higher
8 waves · Jul 30 – Aug 3 UTC



- JUL 30

Galaxy Wave 1 · July 30

+1,082.65 BTC

\$70,406,895
- JUL 31

Galaxy Wave 3 · Jul 31–Aug 1

+208.24 BTC

\$13,542,264
- AUG 1

Morning wave · Aug 1

+0.33295323 BTC

\$21,653
- AUG 1

P2TR wave · Aug 1–3

+46.97389047 BTC

\$3,054,806

- JUL 31

Galaxy Wave 2 · July 31

+76.16 BTC

\$4,952,837
- JUL 31

Evening wave · July 31

+1.20115791 BTC

\$78,114
- AUG 2

Early Aug 2 consolidation

+64.90947964 BTC

\$4,221,193
- AUG 3

Likely Wave 4 · Aug 3

+443.34 BTC

\$28,831,287

BALANCE ON WATCH

1,367.16826524 BTC

\$88,909,687

\$65,032/BTC

STILL HELD

92.0%

of 1,485.7757442 BTC consolidated

MOVED

118.60747896 BTC

LAST MOVEMENT

Block 961,731

Watched holdings

About 1,923.80748125 BTC drained across known waves. Galaxy’s Waves 1–3 same-operator total is 1,367.05 BTC across 4,585 addresses; their Aug 7 high-confidence headline is 1,719.00 BTC across 8,092 addresses (Waves 1–3 + owner-confirmed footprints), or up to ~2,300.00 BTC if outstanding candidates are promoted. 1,485.7757442 BTC is in watched holdings. The remaining ~438.03173705 BTC is mostly unwatched Wave 4 destinations (sparse still-held sample), plus smaller/unmatched Wave 3 vaults, untracked lettered footprints, and fees. Community waves (evening, morning, early Aug 2, P2TR, Wave 4) may be different operators — Galaxy now tracks 25+ attack patterns.

GALAXY WAVE 1 · JULY 30 [Source](#)

Four consolidation holdings. Multi-path key-scan signature (BIP-84/49/44).

SCOPE	BLOCKS	FEE
1,082.65 BTC · 1,196 addresses	960,183–960,191	30 sat/vB · no change 30–75× median that week
WINDOW	PATHS	
July 30, 2026	1,183 BIP-84	
01:10:20 – 01:51:26 UTC	7 BIP-49 · 6 BIP-44	

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
Holding 1	bc1qq85v2c...3fcu9r	562.02139594 BTC \$36,549,375	562.02021083 BTC	57	HELD
Holding 2	bc1qx76cae...htfhe3	398.47576957 BTC \$25,913,676	398.47575357 BTC	4	HELD
Holding 3	bc1q8jy96f...kwtp3q	89.6232989 BTC \$5,828,382	89.6232789 BTC	3	HELD
Holding 4 Unmoved collector remainder	bc1qnk4zh9...8fecp0	32.45060313 BTC \$2,110,328	32.4505632 BTC	164	HELD

GALAXY WAVE 2 · JULY 31 [Source](#)

Two fee bands into a watched vault plus a collector; collector emptied Aug 7 into a hop still holding ~30.18 BTC.

SCOPE	BLOCKS	FEE
76.16 BTC · 1,478 addresses	960,345–960,369	50 and 10 sat/vB

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
Wave 2 vault Consolidated from 1,216 sweeps (via bc1qsjrf5ze...). Chainabuse victim on that sweep sink.	bc1qtfrwa4...numu75	45.90254994 BTC \$2,985,135	45.90251994 BTC	4	HELD

BTC · ~0.69	Aug 1 hop vault → 3KMmeqPe... hub · likely custodial/swap deposit — hops already mixed out; stolen BTC effectively gone from watchable stack
ETH · ~11.7	Evening vault → THOR → 0x6A08... → 0xC61F... → vanity hops → KuCoin 17 · emptied Aug 2 20:29 UTC (~11.70 ETH); consolidated with ~11.08 ETH from a parallel stack into a 23 ETH deposit; also funded via Orbiter Bridge 3 peels
BTC · ~0.27	Evening vault → bc1qdt6c... → bc1qs86u... consol. → KuCoin 328Gx... · KuCoin 3JEq... · emptied Aug 3 01:22 UTC; ~0.27 mixed into a 1.54 BTC multi-input stack (block 960800), then peeled to two KuCoin deposits (~0.70 + ~0.84)

Label	Address	Balance	Report	UTXOs	Status
Evening vault Jul 31 consolidation (~0.51 BTC). Emptied Aug 2–3 (ETH + BTC rails above). Chainabuse victim .	bc1q7rmsw0... lrrcgf	0.00 BTC \$0	0.50980268 BTC	0	EMPTIED
Aug 1 hop vault 0.4998 + 0.1915 BTC stack. Emptied Aug 2 (BTC rail above). Ocean peel also met at the hub (not in report balance).	bc1qayw8nr... 7w2u0s	0.00 BTC \$0	0.69135523 BTC	0	EMPTIED

MORNING WAVE · AUG 1 Source Honey-pot / duress-wallet report. Mk4 claim later withdrawn — seed was Mk3-origin.					
SCOPE	BLOCKS	FEE	PATTERN		
0.33295323 BTC	960,518–960,523	~15–50 sat/vB	direct victim → vault		
Label	Address	Balance	Report	UTXOs	Status
Aug 1 vault	bc1qzm5pau...v34mf3	0.33203236 BTC \$21,593	0.33203236 BTC	16	HELD

EARLY AUG 2 CONSOLIDATION

Source

Distinct fee/structure from Galaxy Waves 1–3. Community Mk2 drain report plus a Mk3 testing-device victim in the same consolidation; sampled inputs long-dormant and funded after the March 2021 vuln window. Chainabuse victim lands in the same vault/tx (block 960668). Likely one of Galaxy’s lettered footprints.

SCOPE	BLOCKS	FEE	WINDOW
64.90947964 BTC · 902 inputs	960,668	~9 sat/vB	04:03 UTC Aug 2
795 addresses			

PATTERN

multi-path → one P2WPKH

AUG 4–5 MIX · WASABI-STYLE CASCADE					
1 · Empty	Aug 2 vault → Taproot hop · ~64.90 BTC · blk 961065 · ~5 sat/vB · e3274a1b...2b3ddf				
2 · CJ1	residual ~54.32 @ bc1qajcrhj...vp2d3t · plus 20× ~1.2914 · blk 961069 · ~2 sat/vB · 324 → 382 · f3ee6e61...c37b24				
3 · CJ2	residual ~47.12 @ bc1qq6s7ws...4jy2fr · equal denoms · blk 961082 · ~2 sat/vB · 320 → 374 · 3bdac8ed...cdf935				
4 · CJ3	residual ends · 7× ~7.7484 (anonymity set) · 2 still live: bc1qf35ycf...puad8k, bc1qs3lj4x...y9v9ze · also 2× ~3.8742 · blk 961093 · ~1 sat/vB · 454 → 502 · 80f11c77...8cbf90				

5 · Remix

other large outs remixed in further CJs (blks 961102–961160) · some equal-denom peels (anonymity set — not proven same operator) hit **Hyperunit**; no labeled CEX on the unique-residual path. Still watching 2× ~7.75 + unique ~3.40 residual

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
Aug 2 vault	bc1q0rvn88...	0.00 BTC	64.90373764 BTC	0	EMPTIED
Emptied Aug 4 → Taproot hop → Wasabi CJ cascade (trail above).	jf5q6m	\$0			

P2TR WAVE · AUG 1–3

Source

Distinct from Wave 4: ~5 sat/vB (later also 25/40) 1-vout multi-path sweeps into Taproot consolidations. Small vault corroborated by a Chainabuse Coldcard-hack victim report; larger sibling/hop verified on-chain with the same fingerprint. Watched set is Kelbie’s published sample, not every P2TR sink.

SCOPE

46.97389047 BTC watched

BLOCKS

960,624–960,897

FEE

~5 sat/vB

PATTERN

1-vout → P2TR

3 Kelbie P2TR sinks

later also ~25–40

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
P2TR vault	bc1pum5zf6...	0.50992489 BTC	0.50992489 BTC	62	HELD
Chainabuse victim · 62× ~5 sat/vB sweeps (blocks 960737–960750).	8ftmy8	\$33,161			
P2TR hop vault	bc1pd133jt...	36.01585057 BTC	36.01585057 BTC	1	HELD
Consolidate from emptied collector bc1ptd5x... (101 inputs @ block 960736).	1uu9ce	\$2,342,183			
P2TR vault	bc1p0l0xs2...	0.00000872 BTC	10.44811501 BTC	2	PARTIAL
Later sibling; fees mostly ~5 sat/vB with 25/40 bands (blocks 960773–960897). Emptied Aug 4 — trail below. ETH side (@osint_based): ~205 ETH via THORChain into 0x41B752... → 200 ETH Tornado (2×100); BTC→ THOR hop soft (commingled hubs). ~4.44 BTC still on Taproot park bc1prjnv....	6q1k0h	\$1			

AUG 4 CASH-OUT · LATER P2TR VAULT (~10.45)

~0.01 → OP_RETURN messenger hub · 961019 · **780ea043...cf0326**

~1.00 → OP_RETURN messenger hub · 961021 · **064eb61d...7ddd4f**

~2.00 → OP_RETURN messenger hub · 961026 · **1a454301...c772c2**

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
~1.00 → OP_RETURN messenger hub · 961032 · 2c7075a3...5012cc					
~2.00 → OP_RETURN messenger hub · 961037 · 92718385...cac6b9					
~4.44 → Taproot park (still held) · 961048 · d969fd1c... 7e16ef					
Then via hop hubs (commingled) → Binance					
~5.33 → Binance deposit · 961034 · 8 peels from bc1q2nfx... (commingled hub) · 0c60a619...5b6850					
~2.43 → Binance deposit · 961046 · from bc1q2nfx... change chain · 57a9d5f2... 7c90bc					
~1.81 → Binance deposit · 961046 · includes vault UTXO taint via prior consolidate · 1c09bf59... af961f					
~5.33 → Binance hot · 961043 · 13i9Za... sweep · e8c70a9d...15ce29					
Parallel ETH rail (@osint_based) · Aug 4 — BTC→THOR hop soft					
~205 ETH → 0x41B752... (via THORChain)					
200 ETH → Tornado.Cash Router · 2×100					

LIKELY WAVE 4 · AUG 3 Source Pattern-match only (no victim report yet). Galaxy’s revised Wave 4 figure is 448.73 BTC after cutting 89 multisig destinations. We go one step further and also drop 6 destinations that already had prior on-chain history (–5.39 BTC), so our total is 443.34. Galaxy’s Aug 7 candidate ceiling is 2,300+ BTC (high-confidence 1,719 plus outstanding promotions, including this Wave 4 figure) — not our stricter 443.34 cut. Watched addresses are a sparse still-held sample. SCOPE 443.34 BTC · 703 addrs BLOCKS 960,778–960,792 FEE ~1–3 sat/vB RBF opt-in PATTERN 1:1 victim → fresh dest likely only — no victim report yet FULL LISTS Confirmed addresses Mempool / RBF list — unfiltered originals					
LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
Wave 4 park	bc1q05g201... nju0ml	8.55600708 BTC \$556,414	8.55600708 BTC	1	HELD

LABEL	ADDRESS	BALANCE	REPORT	UTXOS	STATUS
Wave 4 park	bc1qjenjhm...mv4z23	5.07348539 BTC \$329,939	5.07348539 BTC	1	HELD
Wave 4 park Thorn's unique dual-sweep destination (2 victims → this park). ~2.79 → Bullish.com · 960793 ~2.82 → Coinbase Prime · 960818 · same cluster as bc1qactq...	1N8knQCfjq...WVqfDo	0.00 BTC \$0	5.61303754 BTC	0	EMPTIED
Wave 4 park Emptied Aug 4 (block 961064, ~2 sat/vB): 14-input P2SH consolidate with sibling parks → 334iKwmh... still holding ~136.41 BTC (f02164a9...50db281). Same tx as 342L6n3b... + 36XfMDAYu... + 34nHYNnc... . Mix includes parks funded outside Wave 4 window.	35dHFzKHn4...X4wNxH	0.00 BTC \$0	2.17767618 BTC	0	EMPTIED
Wave 4 park Emptied Aug 4 (block 961064): same 14-input consolidate as 35dHFzKH... → 334iKwmh... (~136.41 BTC still held).	342L6n3b61...TSjZtz	0.00 BTC \$0	2.02374869 BTC	0	EMPTIED
Wave 4 park	bc1q65e6cd...yce9u8	1.14852067 BTC \$74,691	1.14852067 BTC	1	HELD
Wave 4 park Emptied Aug 4 (block 961064): same 14-input consolidate as 35dHFzKH... → 334iKwmh... (~136.41 BTC still held).	36XfMDAYuC...1F1x3G	0.00 BTC \$0	1.09429601 BTC	0	EMPTIED
Wave 4 park	bc1qcmnjt0...rfjvt1	1.04998888 BTC \$68,283	1.04998888 BTC	1	HELD
Wave 4 park	bc1pj7f355...4sulc2	1.03514173 BTC \$67,317	1.03514173 BTC	1	HELD
Wave 4 park Emptied Aug 4 (block 961064): same 14-input consolidate as 35dHFzKH... → 334iKwmh... (~136.41 BTC still held).	34nHYNnc9D...jP4qDQ	0.00 BTC \$0	0.96227191 BTC	0	EMPTIED

Movement feed

Outbound spends of the reported consolidation from watched holdings, then followed destinations (up to hop 2). Stops at known exchange, bridge, and service-hub exits — later custodian churn is not listed. Surplus through a vault that still holds its reported balance, later peels through a hop after it already cashed out to a known exit, and CoinJoin-scale fan-outs are ignored.

HOP 1

a2e65c44...7596bd

-0.0077407 BTC

From Hop 1 from Wave 4 park bc1q72k7...aamt

Block 961,731

To Bybit deposit 17fqFGPG...FcFL

HOP 1

da5064ee...21977f

-0.00308496 BTC

From Hop 1 from Wave 4 park bc1q72k7...aamt

Block 961,730

To bc1q551l...g4q2

HOP 1

cd1d5683...a3fa2b

-0.00768211 BTC

From Hop 1 from Wave 4 park bc1q72k7...aamt

Block 961,598

To bc1q551l...g4q2

HOP 1

18f7ec31...9d561f

-0.00462205 BTC

From Hop 1 from Wave 4 park bc1q72k7...aamt

Block 961,543

To bc1q551l...g4q2

Am I at risk?

Short checklist from Coinkite’s Aug 1 advisory, Block’s root-cause writeup, WizardSardine’s analysis, and Nunchuk’s multisig migration guidance. This tracker does not check your wallet — follow the official steps before moving funds.

LIKELY EXPOSED

Your seed is likely exposed if it was generated on vulnerable firmware. Fixed releases and later are safe for new seeds.

DEVICE	VULNERABLE	FIXED
Mk3	4.0.1–4.1.9, 5.0.1–5.0.3 (~40-bit)	4.2.0+
Mk4 / Mk5	Before Aug 1 patch (~72-bit)	5.6.0+
Q	Before Aug 1 patch (~72-bit)	1.5.0Q+
Edge (Mk4 / Mk5)	Before Aug 1 patch	6.6.0X+
Edge (Q)	Before Aug 1 patch	6.6.0QX+

FIRMWARE UPDATE IS NOT ENOUGH

Patching does not repair an existing seed. Generate a new seed on fixed firmware and migrate funds carefully.

MIGRATE BEFORE YOU UPGRADE

If this Coldcard is your only signer for a weak seed, move funds first. Community reports (including [Jameson Lopp](#)) flag a non-zero chance the hotfix upgrade bricks the unit; an open [firmware PR](#) argues a sticky TRNG fault after the hotfix can lock the keypad before login. Prefer sweeping to a new seed on a second fixed device when possible.

MULTISIG URGENCY & SLIPSTREAM

If Coldcard-generated keys alone could meet your signing threshold (e.g. 2-of-3 with two Coldcard keys), migrate now — those wallets can be swept. One Coldcard key below threshold is lower urgency but still migrate. Broadcasting a migration to the public mempool opens a brief RBF-sniping window while weak keys remain spendable; [Nunchuk](#) recommends submitting the signed hex via [MARA Slipstream](#) instead (now public, no code). That keeps the tx out of the public mempool but trusts MARA and can confirm slower (~5–6% hashrate). No multisig sweeps reported yet.

ORIGIN OF THE SEED

Exposure follows where the seed was generated, not the device holding it today. Seeds imported elsewhere, or whose origin you cannot establish, should be treated as affected.

BEYOND THE MAIN SEED

Weak RNG also hit paper-wallet keys, clone / Key Teleport material, and some Secure Notes passwords — even when the BIP-39 seed itself came from dice ([WizardSardine](#) / [Block](#)).

LOWER RISK

At least 50 private dice rolls at seed creation. A long truly random BIP-39 passphrase (not the device PIN) may buy time, but weak passphrases are not enough — [BTC Sessions](#) reported a confirmed Mk3 drain with a two-word passphrase. [CK tripwire](#) is mapping live how far added dice/passphrase entropy currently stops sweeps (zero-entropy control swept in ~1h; ~5-bit low/dice honeypots swept by Aug 8; mid/high bands still live). Still migrate when practical.

NOT THIS BUG

TAPSIGNER, Opendime, and Satscard use different codebases and are not affected. Mk1 and pre-4.0.1 Mk2/Mk3 firmware used the real TRNG.

SOURCES

Galaxy Research · \$111M update

Aug 7: high-confidence 1,719 BTC / 8,092 addresses (~\$111M); chart established losses 1,719.19. Outstanding candidates → 2,300+ BTC if promoted; total losses likely exceed \$130M. 25+ attack patterns / multiple threat actors. Scope still Coldcard Mk3/Mk4/Mk5/Q post-2021-03-17 firmware. 250+ victim reports via [@intangiblecoins](#).

Galaxy Research · \$100M update

Aug 3: high-confidence 1,596 BTC / ~7,300 addresses across Waves 1–3 + 14 owner-confirmed footprints; candidate Wave 4 would bring ~2,055 BTC. Multiple independent attackers. Waves 1–3 coins then still unmoved (Wave 2 collector later emptied Aug 7). Superseded by the Aug 7 \$111M cut.

Alex Thorn · Footprint O

Aug 4: ≥15 attackers (footprints A–O). Overnight footprint O ~12 BTC from 126 addresses from one victim report; later folded into Galaxy's Aug 7 confirmed footprint set (no public address list).

James O'Beirne · CK tripwire

Aug 4: live honeypot scoreboard mapping attacker brute-force depth via decoys with added dice/passphrase entropy ([cktripwire.com](#); [announcement](#)). Zero-entropy control swept in ~1h. By Aug 8, multiple ~5-bit low (dice) honeypots were swept ([jamesob](#)); frontier still ~5 bits with mid/high passphrase bands live. Defensive research only — do not add honeypot sinks to holdings.

Galaxy Research · Wave 3

Aug 1: third wave (~207.73 BTC into 293 P2WSH vaults). Observed total 1,367.05 BTC across 4,585 addresses.

Galaxy Research · Wave 2

Aug 1: second wave attributed to the same operator — 1,158.81 BTC / 2,673 addresses across seven holdings.

Galaxy Research · Wave 1

Mapped the July 30 fingerprint set: 1,082.65 BTC across ~1,195 addresses into four holdings.

Kevin Kelbie

Independently flagged the ~45.9 BTC July 31 vault (now Galaxy Wave 2) after Block's scan window; RBF differed from Wave 1.

Kevin Kelbie · P2TR wave

Aug 3: ~0.51 BTC Chainabuse-corroborated P2TR sink plus two larger sibling Taproot consolidations (~36 + ~10 BTC) at ~5 sat/vB — distinct from Wave 4.

@osint_based · P2TR later-vault ETH rail

Aug 5: later P2TR vault cash-out — ~205 ETH via THORChain into 0x41B752... → 200 ETH Tornado Cash (2×100). Same Kelbie later vault (bc1p0l0...), not a new cluster; BTC → THOR amount/path soft through commingled hubs. ~4.44 BTC still parked on bc1prjnv....

Evan Schoenberg

Evening July 31 low-fee (~2 sat/vB) sweeps into a separate vault; Aug 1 hop consolidation (incl. delayed ~0.19 BTC) raised the watched cluster to ~1.20 BTC.

Ocean block 960511

Coinbase of block 960511 (tag OCEAN.XYZ) paid ~0.060 BTC that later touched the Aug 1 hop vault and was peeled off (not counted as stolen). That peel later hit the same P2SH service hub (**3KMmeqPe...**) as the stolen evening-wave hops — same exit venue, not proof of same operator.

Tomer Strolight

Aug 1 morning wave: duress-wallet honey pot swept into a ~0.33 BTC vault; Mk4 attribution later corrected to Mk3-origin seed.

Alex Thorn · Likely Wave 4

Aug 3: likely 4th wave by pattern match (no victim report yet). Filtered core ~443.34 BTC / 703 addrs after removing 89 multisigs (~20.58 BTC, per Nunchuk) and 6 prior-history destinations (~5.39 BTC). Original pastebins still list the unfiltered set.

Marius Offchain

Early Aug 2 consolidation (04:03 UTC): ~65 BTC from ~795 addresses into one vault (community Mk2 drain report); not a Galaxy-published wave.

Erik

Mk3 testing-device victim (~9k sats, native segwit) swept in the Early Aug 2 consolidation — same tx / vault as the Marius report.

Coinkite advisory

Official guidance (updated Aug 1): who is exposed, fixed firmware versions, and migration steps.

Block Engineering

Root-cause writeup on the RNG fallback, 32-bit reseed, and blast radius beyond BIP-39 seeds.

WizardSardine

Aug 1 deep dive: who is affected (including Mk3 5.0.x), advanced features still broken with dice seeds, and migration guidance.

Nunchuk · Slipstream migration

Aug 3: multisig urgency tiers (Coldcard keys alone vs below threshold) and MARA Slipstream to avoid RBF-sniping during migration; Slipstream now public with no code.

Kevin Loaec

Announcement thread for the WizardSardine writeup.

CoinDesk

Aug 1 reporting on Galaxy's expanded totals and why further waves remain possible.

Clay Garrett

July 30 operator used a paid blockchain-data account during the sweeps; findings shared with authorities.

